

with gender-neutral usability and actively promoted to women entrepreneurs is crucial.

Additionally, encouraging and measuring the participation of women-led businesses in national digital transformation initiatives (e.g., Saudi's LEAP conference, Dubai's Future District) can be a key performance indicator for inclusivity. Ultimately, the goal of measurement is to enable a transition from bridging the digital divide to fostering digital advantage for GCC women entrepreneurs, ensuring they are not only participants but also leaders in the region's digital economic future.

Conclusion

Measuring the gender digital divide in GCC entrepreneurship requires a nuanced, layered approach that distinguishes between having digital tools and wielding them effectively for business growth.

While the region exhibits favourable conditions in terms of basic digital infrastructure, significant gaps persist in the adoption of strategic, business-enhancing technologies and in the conversion of digital access into scalable entrepreneurial outcomes. A comprehensive measurement framework that integrates data on access, adoption, and effective utilization-while accounting for sectoral segmentation and socio-cultural dynamics-provides the evidence base necessary for impactful action.

For the GCC nations to fully realize the dual goals of their national visions-digital leadership and empowered female economic participation-a concerted effort to diagnose and bridge this strategic divide is not merely advantageous but essential.

It is through this lens that digitalization can truly fulfill its promise as a democratizing force, enabling a new generation of women entrepreneurs to build competitive, innovative, and resilient enterprises that contribute to the region's diversified and sustainable economic prosperity.

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PLATFORM WORK AND SOCIAL PROTECTION GAPS IN ASEAN'S GIG ECONOMY

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The explosive growth of digital labour platforms across the Association of Southeast Asian Nations (ASEAN) has redefined work for millions, offering flexibility and income opportunities in economies characterized by high informality. However, this article argues that the platform-mediated gig economy is systematically generating and exacerbating profound gaps in social protection for its workers. Through an analysis of sectors including ride-hailing, delivery, and micro-tasking, this paper examines the misclassification of platform workers as independent contractors rather than employees, which legally severs their access to nationally mandated social security schemes covering health, work injury, pensions, and unemployment. The analysis identifies a tripartite crisis of coverage, financing, and portability. While platform companies optimize for flexibility and minimal liability, workers bear disproportionate risks of occupational hazards, income volatility, and old-age poverty. The region's diverse regulatory responses-from proactive judicial rulings and pilot schemes in Thailand and Malaysia to more laissez-faire approaches in Indonesia and the Philippines-are mapped and assessed. The paper concludes that the current trajectory entrenches a new, digitally-enabled precariat. It advocates for innovative policy models that delink core protections from strict employment contracts, proposing mechanisms such as platform levies to fund portable benefits, the establishment of earnings-based contribution systems, and regional dialogue to establish minimum standards, thereby ensuring that economic digitalization does not come at the cost of social regression.

Keywords: platform work, gig economy, social protection, ASEAN, digital labour, worker classification, portable benefits, precarity, ride-hailing, social security

Introduction

The ASEAN region, with its young population, high mobile penetration, and burgeoning urban middle class, has become a fertile ground for the expansion of the platform-based gig economy. Companies like Grab,

Gojek, Foodpanda, and numerous local micro-task platforms have created a vast, on-demand labour market.

This model provides crucial income generation, particularly for youth, migrants, and those transitioning from agrarian or informal sectors. Yet, the very architecture of platform work—built on algorithmic management, performance ratings, and the formal legal designation of workers as independent partners or contractors—fundamentally disrupts the traditional employer-employee relationship that has long served as the primary gateway to social protection.

This article investigates the resulting social protection gaps that define the working lives of ASEAN's platform workers. It contends that these gaps are not incidental but are embedded in the business models of digital labour platforms, raising urgent questions about social equity, risk pooling, and the sustainability of work in an increasingly digitalized ASEAN economy.

The architecture of precarity: business models and legal classification

The social protection deficit originates in the foundational legal and operational design of platform work. Platforms position themselves as mere technology intermediaries connecting service providers (drivers, delivery riders, freelancers) with customers, thereby denying statutory employer status.

This classification is pivotal, as labour laws and social security systems across ASEAN—such as Indonesia's BPJS Ketenagakerjaan, Thailand's Social Security Fund, and the Philippines' SSS—typically mandate contributions for employees, funded jointly by employer and employee. As non-employees, platform workers are excluded from these schemes. Their engagement is governed by unilaterally set terms of service that grant platforms maximum discretion over incentives, penalties (via deactivation), and fee structures, while assuming minimal responsibility for worker welfare.

Algorithmic management further obscures accountability, making it difficult to attribute occupational risks to a specific "employer." Consequently, workers assume all entrepreneurial risk—from fluctuating demand and vehicle maintenance to accident costs—without enjoying the corresponding entrepreneurial upside or safety net.

Mapping the social protection gaps

The exclusion from formal employment-based systems manifests in critical, interdependent gaps across the core branches of social security.

This is the most acute gap. Platform work, particularly in transportation and delivery, carries significant risks of traffic accidents. Without employer-linked health insurance or workers' compensation, injured riders face catastrophic out-of-pocket medical expenses and loss of income during recovery. While some platforms offer ad-hoc commercial insurance

packages, these are often limited in scope, duration, and value, requiring workers to pay premiums and navigate complex claims processes.

The irregular and piece-rate nature of platform income discourages voluntary, long-term savings. Most platform workers do not contribute to any pension scheme, storing up a crisis of old-age poverty for a generation that will have no agricultural land to fall back on and insufficient savings. The absence of employer contributions makes voluntary schemes prohibitively expensive for low-earning gig workers.

Platform work is intrinsically volatile, subject to algorithmically determined demand surges and slumps, weather conditions, and platform competition. There is no protection against income loss due to illness, platform deactivation, or market downturn. The "flexibility" marketed by platforms translates into a complete individualization of unemployment risk.

The legal status of independent contractors typically denies platform workers the right to form unions or engage in collective bargaining, stripping them of a key mechanism to negotiate for better earnings, safety standards, and grievance redress, further entrenching their vulnerability.

Table 1 - Typology of social protection gaps for platform workers in ASEAN

Protection Type	Platform Worker Reality	Consequence
Health & Work Injury	Reliance on personal insurance or limited, opt-in platform schemes with high co-pays.	Medical debt, impoverishment from accidents, delayed return to work.
Pension/ Old-Age	No employer contribution; voluntary savings unlikely due to income variability.	High risk of old-age poverty and dependency.
Income Security	No income when not working (illness, deactivation, low demand).	High financial stress, inability to plan, forced overwork.
Collective Voice	Classified as independent businesses; collective action often challenged legally.	Lack of leverage to improve terms, conditions, or safety standards.

Regulatory responses and innovative experiments in ASEAN

ASEAN member states exhibit a spectrum of regulatory approaches, reflecting divergent political economies and labour market structures.

Proactive Judicial and Legislative Action: Thailand's Social Security Court has ruled in specific cases that certain platform workers should be considered employees under the Social Security Act, granting them access to benefits. Malaysia has implemented the Self-Employment Social Security Scheme (SKSPS) under the SOCSO, a voluntary but state-subsidized scheme for gig workers, representing a pioneering step towards delinking protection from employment status.

Platform-Facilitated, Limited Schemes: In Indonesia and the Philippines, the dominant approach has been to encourage platforms to offer voluntary insurance products. For example, platforms partner with insurers to offer accident coverage, but these remain fragmented, non-portable, and often inadequate. The Philippine Social Security System (SSS) has a voluntary program for self-employed, but uptake among gig workers is low due to cost and awareness.

Regional Dialogue and Soft Law: ASEAN as an institution has begun discussions on the future of work, but binding regional standards on social protection for platform workers remain absent. The challenge is harmonizing responses across vastly different national systems while preventing a regulatory race to the bottom.

Policy pathways towards inclusive protection

Closing the protection gap requires moving beyond the rigid employee/contractor binary. Policy innovation should focus on creating portable, prorated, and universal benefits.

Establish earning-based contribution systems where obligations are tied to work activity, not contractual form. A small levy on each transaction or a percentage of platform fee could be directed into dedicated social security funds, with contributions split between the platform, the user (as client), and the worker.

Workers could accumulate rights in individual portable benefit accounts linked to their digital identity. Benefits would follow them across multiple platforms and gigs, covering health, injury, and pension savings.

Governments must explicitly extend core protections, starting with work injury insurance, to all workers, irrespective of classification. This can be financed through a platform levy or a broader digital services tax.

Creating formal tripartite-plus forums (government, platforms, worker associations, civil society) is essential to design feasible and fair solutions, including setting minimum earning standards and transparent deactivation procedures.

Conclusion

The rise of platform work in ASEAN has laid bare the inadequacy of 20th-century social protection models in a 21st-century digital labour market. The current system, which permits platforms to externalize the social costs of work, is creating a vast, vulnerable workforce without safety nets. While the gig economy offers valued flexibility, it must not become a synonym for heightened precarity and abandoned social responsibility.

The diverse experiments within ASEAN, from Thailand's judicial interventions to Malaysia's portable scheme, provide a foundation for learning. The imperative now is for bolder, systemic innovation that recognizes work wherever it occurs and ensures that all workers, regardless of how they are engaged, contribute to and benefit from collective risk-pooling mechanisms.

The future of social cohesion and equitable growth in ASEAN depends on building an inclusive digital social contract that leaves no worker behind. Achieving this requires political will to recalibrate the balance between platform innovation and worker security, ensuring the region's digital transformation is both economically dynamic and socially sustainable.

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DIGITAL SKILLS AS A DRIVER FOR YOUTH INCLUSION IN MOUNTAIN ECONOMIES

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Mountain economies, characterized by geographic isolation, challenging terrain, and often limited traditional economic opportunities, face significant risks of youth outmigration and demographic decline. This article posits that the strategic development of digital skills among youth populations represents a critical, non-geographically bound driver for economic and social inclusion, capable of unlocking new livelihood pathways and revitalizing these communities. Moving beyond infrastructural debates around broadband access, the analysis focuses on the transformative agency conferred by competencies in digital literacy, remote collaboration, e-commerce, digital marketing, and niche technical skills such as GIS mapping and data analysis. These skills enable youth to participate in the global digital economy, create location-independent enterprises, and add value to traditional mountain sectors like sustainable tourism, agroforestry, and artisanal production. However, the paper identifies a dual challenge: first, the frequent misalignment of generic digital literacy programs with the specific